



BIOLOGY CH.2 — NUTRITION IN ANIMAL — FULL MIND MAP

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1. DNA — Genetic Information Source

- DNA (Deoxyribonucleic acid) living organisms mein protein formation ke liye genetic information ka source hai — ye 2 polynucleotide chains ka double helix polymer hai, jiska code 4 bases (Adenine, Guanine, Cytosine, Thymine) se bana hota hai. [Q9]

2. Nutrition in Amoeba

- Amoeba mein, 'Pseudopodia' (finger-like projections) food lene mein help karte hain, aur 'Food vacuole' digestion of food karta hai — Amoeba pond water mein milne wala single-celled organism hai jo apna shape-position constantly badalta rehta hai. [Q10]

3. Body-Building Foods — Proteins & Pulses

- Protein-rich food (jaise Peas) ko 'body-building food' kehte hain — protein amino acids ki long chains se bana hai, DNA molecules regenerate karne, hormones (Insulin, Secretin) banane, aur immune system mein help karta hai. Sources: Beans, Pulses, Yogurt, Fish, Soya, Nuts. [Q11]
- Pulses growth ke liye proteins provide karti hain — sasti protein-vitamins-fiber ka source hain. Protein deficiency se Kwashiorkor aur Marasmus jaisi diseases hoti hain. [Q14]

4. Autotrophs — Starch Storage

- Autotrophs apna food 'Starch' ke roop mein store karte hain — din mein photosynthesis se starch banta hai, raat mein degrade hokar leaf metabolism support karta hai (jaise Leaf mesophyll mein). [Q12]

5. Heterotrophic Organisms

- Heterotrophs doosre plants-animals khaakar energy-nutrients paate hain — Tiger ek heterotrophic organism hai (Algae, Grass, Neem tree autotrophic hain). [Q13]

6. Balanced Diet — Milk

- Milk apne aap mein ek balanced food hai — Calcium, Riboflavin, Phosphorus, Vitamins A-B12, Potassium, Magnesium, Zinc, Iodine, lactic acid deta hai. India mein average calorie requirement rural areas mein 2400 aur urban areas mein 2100 calories/din hai. [Q15]

7. Vitamin C — Sources & Additive Code E300

- Citrus fruits Vitamin C ka richest source hain (Kiwifruit, Orange, Broccoli, Strawberry, Mango, Lemon, Papaya bhi shaamil). [Q16]
- 'E300' ka common naam Vitamin C (Ascorbic acid) hai — ek water-soluble vitamin jo bodily-immune functions mein crucial role play karta hai. [Q20]

8. Energy from Food — Fats Provide the Most

- Fats per gram sabse zyada energy dete hain (9 kcal/g) — carbohydrates-proteins lagbhag 4 kcal/g dete hain, aur Roughage (fiber) koi energy nahi deti. [Q17]
- Fats breakdown hone par sabse zyada energy dete hain — 9 calories/gram, protein-carbohydrates se double se zyada. [Q28]

9. Vitamin B12 — Cyanocobalamin & Cobalt

- 'Cyanocobalamin' Vitamin B12 ka man-made form hai — central nervous system development, red blood cell formation aur DNA synthesis ke liye zaroori. Naturally animal-origin foods (fish, meat, poultry, eggs, dairy) mein milti hai. [Q18]
- Cobalt element Vitamin B12 mein present hota hai — meat, fish, eggs aur dairy products mein main source hai. [Q23]

10. Vitamin A — Active Forms & Beta Carotene

- Vitamin A ke 3 active forms hote hain: retinal, retinol, aur retinoic acid — Casimir Funk ne vitamins discover kiye. Vitamin A source: Nuts, Eggs, Fish, Liver, Green leafy vegetables; deficiency se Night Blindness, Dry Skin-Eyes hoti hai. [Q19]
- Beta Carotene (red-orange pigment, carrots-colourful vegetables mein) body mein 'Vitamin A' mein convert hota hai. [Q36]

11. Saprotrophic (Saprophytic) Nutrition

- Saprotrophic nutrition mein organism dead-decaying matter se food leta hai — fungi aur soil bacteria isse follow karte hain, ye ek chemoheterotrophic extracellular digestion process hai. [Q21]
- Organisms jo dead-decaying matter se solution form mein nutrients lete hain, unhe 'Saprophytic nutrition' kehte hain — example: Mucor, yeast. [Q24]

12. Iodine — The Thyroid Micronutrient

- Iodine, thyroid gland ke normal functioning ke liye essential micronutrient hai — thyroid cells food se iodine absorb karke thyroxine (T4)-triiodothyronine (T3) hormones banate hain. Iron deficiency se Anemia, Zinc se skin disease, Calcium se soft bones hoti hai. [Q22]

13. Glycogen — Carbohydrate Storage Form

- Glycogen body mein carbohydrates ki storage form hai — excess glucose glycogen mein convert hokar liver-muscles mein store hota hai, energy zaroorat par wapas glucose mein todta hai. [Q25]

14. Casein — Milk's White Protein

- Casein ek protein hai jo Milk mein paya jaata hai — milk ka white colour Casein ki wajah se hota hai (cow's milk mein ~80% casein protein hoti hai). [Q26]

15. Protein Denaturation

- ⚠️ Trap: 'Proteins temperature se affected nahi hote' ye statement GALAT hai — proteins low temperature mein inactive ho jaate hain, high temperature aur pH change se denature ho jaate hain (secondary/tertiary structure disturb hoti hai). Types: Fibrous proteins (keratin, myosin) aur Globular proteins (Insulin, albumins). [Q27]

16. Keratin — The Structural Material of Hair

- Keratin hair ka main structural material hai — ek fibrous protein jo epidermis mein milta hai, skin surface ko strength aur water-resistance deta hai. [Q29]

17. Myosin & Actin — Muscle Proteins

- Myosin aur Actin — dono muscle proteins hain. Actin globular multi-functional proteins ka family hai jo cytoskeleton mein microfilaments banate hain — Myosin chemical energy (ATP) ko mechanical energy mein convert karke force-movement generate karta hai. [Q30]

18. Vitamin B1 — Destroyed by Heat

- Vitamin B1 (aur baaki water-soluble vitamins — B-complex, C) heat se destroy ho jaate hain, cooking water mein leach ho jaate hain — Fat-soluble vitamins (A, D, E, K) heat-stable hote hain. [Q31]

19. Elements in Proteins — Iodine is NOT One

- ⚠️ Trap: Iodine proteins ka part NAHI hai — proteins mein Carbon, Hydrogen, Oxygen, Nitrogen aur Sulphur hote hain. Iodine thyroid hormones ka essential component hai, iodized salt-seafood mein milta hai. [Q32]

20. Fish Liver Oil — Rich in Vitamins A & D

- Fish liver oil (jaise Cod liver oil) Vitamin D AUR Vitamin A — dono mein rich hoti hai. Vitamin A (Eggs, milk, carrots, mangoes), Vitamin D (fatty fish, fish liver oils, fortified milk-cereals) mein milta hai. [Q33]

21. Vitamin K — Produced by Intestinal Bacteria

- Vitamin K, Intestine mein bacteria dwara naturally produce hoti hai — fat-soluble vitamin, green leafy vegetables mein present hoti hai, blood clotting aur bone health mein help karti hai. [Q34]

22. Lactose — A Disaccharide

- Lactose (milk sugar, $C_{12}H_{22}O_{11}$) Glucose aur Galactose se milkar bani disaccharide hai — 2 monosaccharides ek glycosidic bond se jude hote hain. [Q35]

23. Glucose — Cellular Energy Source

- Glucose body ke cells mein O₂ ke saath combine hokar energy provide karta hai — carbohydrates digestion se glucose banta hai, respiration reaction: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy}$. [Q37]

24. Placenta — Embryo's Nutrition Source

- Embryo, mother ke blood se nutrition 'Placenta' (special tissue, uterus mein pregnancy ke dauran develop hoti hai) ke through leta hai — oxygen-nutrients deta hai, waste products remove karta hai. [Q38]